7 A circuit consists of an a.c. supply and two lamps. The lamps are connected in parallel.

(a) Draw the circuit diagram.

[2]

(b) The electromotive force (e.m.f.) of the a.c. supply is 230 V. When connected to the 230 V supply, the resistance of one lamp is 1200 Ω and the resistance of the other lamp is 800 Ω.

(i) Calculate the current in the 800 Ω lamp.

current = ... [2]

(ii) Calculate the energy used by the 800 Ω lamp in 5.0 hours.

Give your answer in kWh.

energy = ... kWh [3]

(iii) Calculate the combined resistance of the two lamps in this circuit.

resistance = ... [2]

[Total: 9]

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